Research Project

Targeted v.s. Universal Fiscal Policy: A HANK Approach

1. Introduction and Motivation

Fiscal transfers have been an important public policy instrument in various governments. Among the main examples, we can cite: *Bolsa Família* (Brazil), *Asignación Universal por Hijo* (Argentina), *Progresa/Oportunidades* (Mexico), and *Universal Credit* (United Kingdom). Although each program has its specificities, they all share the same objective: income redistribution.

Whether a fiscal policy is targeted or universal makes all the difference when the issue is social justice. In standard macroeconomic models, all individuals are equals, meaning that transfers affect everyone in the same way (Blanchard and Galí 2007).

However, households differ dramatically in their income, wealth, employment status, and propensity to consume. Microdata show that individuals have different marginal propensities to consume (MPC) depending on their income and credit constraints (Kaplan and Violante 2014).

Therefore, a heterogeneous agent model is crucial to understand the impacts and differences between targeted and universal fiscal policies. In Heterogeneous Agent New-Keynesian (HANK) models, these differences matter a lot: the same transfer budget produces very different aggregate and distributional outcomes depending on targeting (Auclert, Rognlie, and Straub 2025; Kaplan and Violante 2014).

Literature has studied the effects of police targeting primarily in relation to police funding: whether through debt or taxes (McKay and Reis 2016). Some other articles have studied the presence of fiscal transfers and unemployment insurance, mostly in isolation (Krueger, Mitman, and Perri 2016; Bartal and Becard 2024).

The idea behind this project is to systematically compare the effects of different fiscal policies to a universal lump-sum transfer, in the presence of a negative shock to aggregate demand that triggers the transfers. To answer this, I aim to simulate different fiscal structures, such as transfers based on wealth, income, employment status, or hand-to-mouth status, in a structural HANK model with unemployment.

2. Model and Methods

2.1 Model

Throughout this project, in the future, I intend to solve, calibrate, and estimate a two-asset HANK model with unemployment and different transfer design, an extension to Kaplan, Moll, and Violante (2018) and Auclert, Bardóczy, and Rognlie (2023). Nevertheless, due to time constraint, this project was carried out with a single asset with labor-search (Gornemann, Kuester, and Nakajima 2016), using unemployment insurance transfer.

Consider a simple incomplete markets model with unemployment risk and one liquid asset. We assume an Aiyagari (1994)-style household environment, where households can be either employed or unemployed, $s \in \{E, U\}$, exogenously assigned by the Probability Matrix Π_t^s, where the employed can lose their job with probability σ_t and the unemployed can find a job with probability f_t, which are endogenously determined by the search-and-match conditions. Thus, the steady-state unemployment is:

$$U = \frac{s}{s+f}$$

Employed workers earn wage $w_t e$, while unemployed receive benefits b_t. Let $y_t(s,e)$ be this non-financial income. Utility is a CRRA function of only consumption, which means we assume that the supply of working hours is inelastic. The Bellman Equation for each individual i is:

$$\begin{aligned} V_t(s,e,a) = &\max_{c,a'} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}_t \left[V_{t+1}(s',e',a'|s,e) \right] \right\} \\ \text{s.t.} \quad & c + a' = (1+r_t)a + y_t(s,e) + T_t \\ & a' \in [\underline{a}, \infty), \end{aligned}$$

where

$$y_t(s,e) = \begin{cases} (1-\tau_t) w_t e, & \text{if employed} \\ b, & \text{if unemployed.} \end{cases}$$

We solved the household problem using the Endogenous Grid Method (EGM) (Carroll 2006), determining the marginal value of the assets V_a, which is:

$$V_a = \frac{\partial V_{t+1}}{\partial a'} = \frac{\partial V}{\partial c} \cdot \frac{\partial c}{\partial a} = (1+r_{t+1}) c^{-\sigma}.$$

The Envelope Condition tells us:

$$u'(c_t) = c_t^{-\sigma} = \beta \mathbb{E}_t [V_a^{t+1}] = \beta \mathbb{E}_t \left[(1+r_{t+1}) c_{t+1}^{-\sigma} \right].$$

Thus, we can create an inverse backwards iteration function of V_a^t with respect to V_a^{t+1}, which is the central idea of the EGM.

In the firm side, there is a representative final-good producer that aggregates a continuum of intermediate inputs by $j \in [0,1]$, $Y_t = \left(\int_0^1 Y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}}\, dj\right)^{\frac{\varepsilon}{\varepsilon-1}}$, where $\varepsilon > 0$. This Constant Substitution Elasticity (CES) function will be incorporated in the future, but for now it serves to derive the Phillips Curve.

Each intermediate good j is produced by a monopolistic competitive firm $Y_{j,t} = Z_{j,t}N_{j,t}$. They hire labor at wage w_t and define a markup of μ.

Each producer chooses price to maximize profit subject to a quadratic price adjustment cost as in Rotemberg (1982), but in discrete-time terms. The New Keynesian Phillips Curve (NKPC) characterizes the inflation $\pi_t = P_t/P_{t-1} - 1$ conditional on (Y_t, Z_t, w_t, r_t):

$$\log(1+\pi_t) = \kappa\left(\frac{w_t}{Z_t} - \frac{1}{\mu}\right) + \frac{1}{1+r_{t+1}}\frac{Y_{t+1}}{Y_t}\log(1+\pi_{t+1}).$$

Monetary Policy follows a standard lagged Taylor Rule, without microfoundations and only with an inflation target, $i_t = r^*_{t-1} + \phi\pi_{t-1}$, where r^* is the neutral real interest rate, $\phi > 1$, and nominal rates follow the Fisher equation $(1+i_{t-1}) = (1+r_t)(1+\pi_t)$ (**woodford2001taylor**). This assumption is very common in New Keynesian models and fits the data very well.

In the governmental sphere, we have a government that only collects taxes, makes transfers, and incurs debt; therefore, it does not spend money, $G_t = 0$. The government's budget constraint is characterized by

$$b_tU_t + T_t + (1+r_{t-1})B_{t-1} = B_t + \tau_t w_t L_t,$$

which means that, in steady-state, $T = \tau wL - rB - bU$. For dynamic simulations, we intend to apply two regimes: one financed by debt, with T fixed and $B_t = (1+r)B + T + b(1-L) - \tau wL$, and another financed by taxes, with B fixed and T variables. For now, we consider a T_t targeted in zero (no lump-sum transfers for the time being) and b_t as exogenous, to calculate the IRFs based on it.

Finally, the asset, labor, and goods markets reach equilibrium (Market Clearing Conditions). We define $\Lambda_t(s,e,a)$ as the household distribution over a given period of time. First, the asset market reaches equilibrium when household savings equal government bonds

$$A_t = \int a_{i,t}\, d\Lambda_t(s,e,a) = B_t;$$

the labor market clears when the share of employment in the workforce equals the

firm's demand for labor

$$L_t = \int \mathbb{1}\{s_{i,t} = E\}\, d\Lambda_t(s, e, a) = 1 - U_t;$$

and the goods market clear when $Y = C$ in the steady (because $\pi \approx 0$), due to the no government spending and no investment assumptions, but, due to the Rottemberg's adjustment cost, outside the steady-state it clears when:

$$\left[1 - \frac{\mu}{\mu - 1}\frac{1}{2\kappa} \cdot [\log(1 + \pi_t)]^2\right] Y_t = (1 - \psi(\pi_t)) \cdot \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}}\, dj\right)^{\frac{\varepsilon}{\varepsilon-1}} = \int c_{i,t}\, d\Lambda_t(s, e, a) = C_t.$$

For simplicity, we only consider the approximations around the steady state, which makes sense because the computational method implemented for the Jacobians linearizes the equations around the steady state.

2.2 Computational Methods

The main method is the Sequence-Space Jacobian (SSJ) (Auclert, Bardóczy, Rognlie, and Straub 2021), which can be used in a `Python` library. The central idea is to summarize the model by a matrix equation $\mathbf{F}_t(\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, where $\mathbf{X}$ are the endogenous variables and $\mathbf{Z}$ are the exogenous variables.

The method is built to find General Equilibrium Jacobians, written as

$$d\mathbf{X} = \mathbf{G}\, d\mathbf{Z} = -\mathbf{F}_{\mathbf{X}}^{-1}\mathbf{F}_{\mathbf{Z}}\, d\mathbf{Z}.$$

We are interested in calculating $dC = \mathbf{G}_{C,b}\, db$, the total effect of the change in unemployment insurance on household consumption. To do this, we divide the model into blocks so that the households are heterogeneous.

The main method for resolving heterogeneity is the Endogenous Grid Method (EGM) (Carroll 2006), which solves the value function iteration using the Euler equation, without solving the family maximization problem.

We assume a Rouwenhorst Markov Process for income $\log(y_t) = \rho_e \log(y_{t-1}) + \varepsilon_t^e$, with $\varepsilon_t^e \sim N(0, \sigma_e^2)$, which is more accurate than Taucher for highly persistent processes. We also assume a log-spaced grid for assets, in order to generate more robustness near the credit constraint.

We solve the steady-state equilibrium computationally by Brent's method (Brent 1973), calibrating the parameters $(\sigma, f, s, \rho_e, \sigma_e, n_E, a_{min}, a_{max}, n_A, Z, \pi, r, \tau, b, B, \phi, r^*, \mu, \kappa)$, and estimating (β, w, Y) aiming at clearing the asset and goods markets and the NKPC. The labor market was left without defined criteria, as were U and the labor market, for the purposes of sanity checks, and everything went well.

3. Results

We solve the model at the calibrated steady state and compute both partial-equilibrium (PE) and general-equilibrium (GE) Jacobians with respect to the unemployment benefit b, treating it as the employment-targeted fiscal instrument. The universal lump-sum transfer is set to zero for now ($T_t = 0$), so that the benefit shock is the sole fiscal intervention. The calibrated and estimated values are in the appendix.

Steady-state distribution. The stationary wealth distribution exhibits a sharp concentration near the borrowing constraint. Approximately 0.5% of households hold zero liquid assets and are therefore hand-to-mouth (HtM), while the Lorenz curve confirms substantial wealth inequality: the bottom 50% of the population holds a near-zero share of aggregate wealth. The steady-state unemployment rate is $U^* = s/(s+f) = 20\%$, consistent with the theoretical target. These two features, a non-trivial HtM share and a sizeable pool of unemployed households, are the key distributional facts that make fiscal targeting welfare-relevant in this economy.

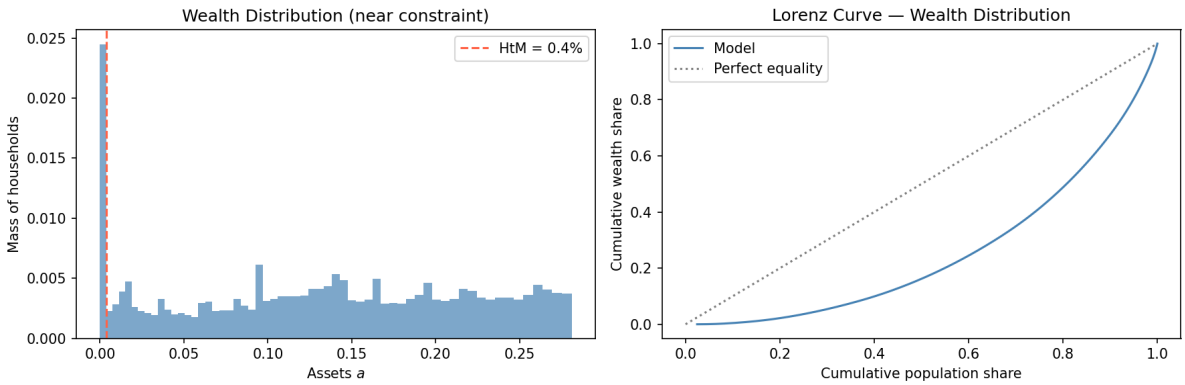

Figure 1: Wealth Distribution

Consumption policy function. The consumption policy function is nearly linear in assets for both employed and unemployed households at the median productivity level. The two schedules are nearly parallel, with the unemployed consuming slightly less at every asset level due to the lower benefit income $b < (1-\tau)w\bar{e}$. The slope of both curves is close to the real interest rate r, indicating that high-asset households are close to permanent-income consumers and will save most of any windfall, a finding consistent with Bartal and Becard (2024) and central to the targeting argument.

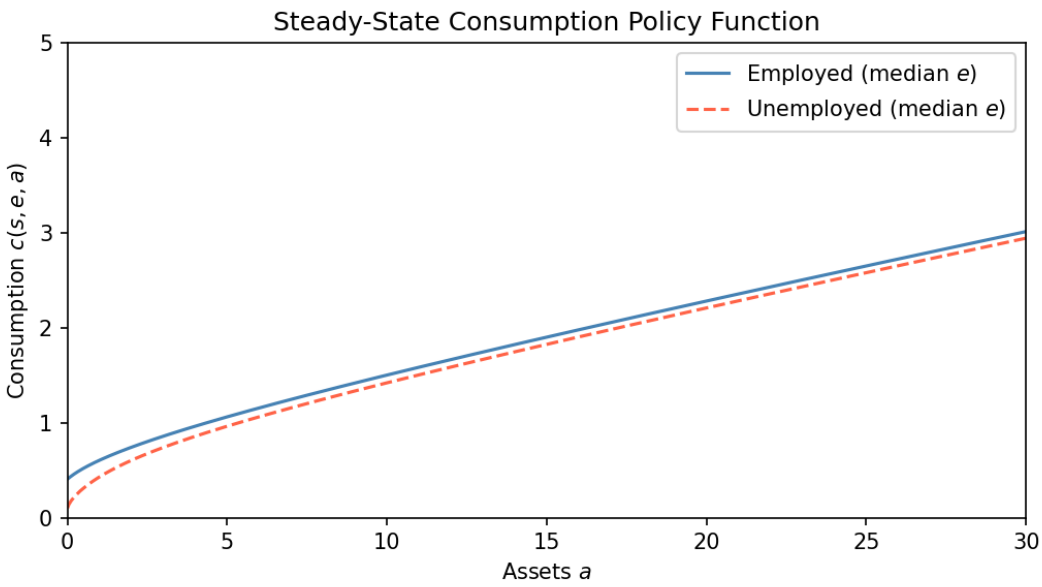

Figure 2: Steady-State Consumption Policy Function

Intertemporal MPC profile. The PE intertemporal MPC of aggregate consumption with respect to b decays rapidly from an impact value of 0.059 at $t = 0$ to near zero by $t = 15$. This profile reflects the fact that unemployed households are disproportionately constrained and spend their additional income quickly. The monotone decay, without the hump shape typical of models with patient savers, is consistent with the one-asset structure: there is still no liquid-illiquid wedge that would delay consumption.

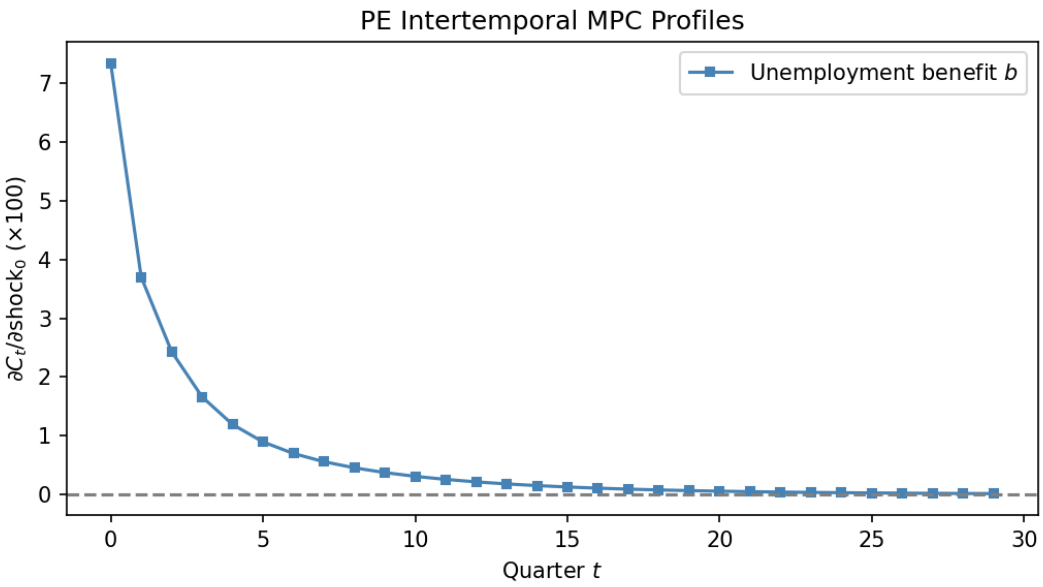

Figure 3: Partial Equilibrium MPC Profiles

General equilibrium impulse responses. We compute GE impulse responses to a persistent 1% rise in b (AR(1) with $\rho = 0.4$). The GE output and consumption multipliers coincide with the PE impact at 0.059, indicating that GE amplification through wage and interest-rate channels is negligible for this shock at the current calibration. The unemployment rate IRF is essentially zero, as expected: since employment transitions are exogenous in this model, a benefit shock cannot directly affect job-finding or separation rates, and the GE wage response is too small to generate meaningful labor-market feedback.

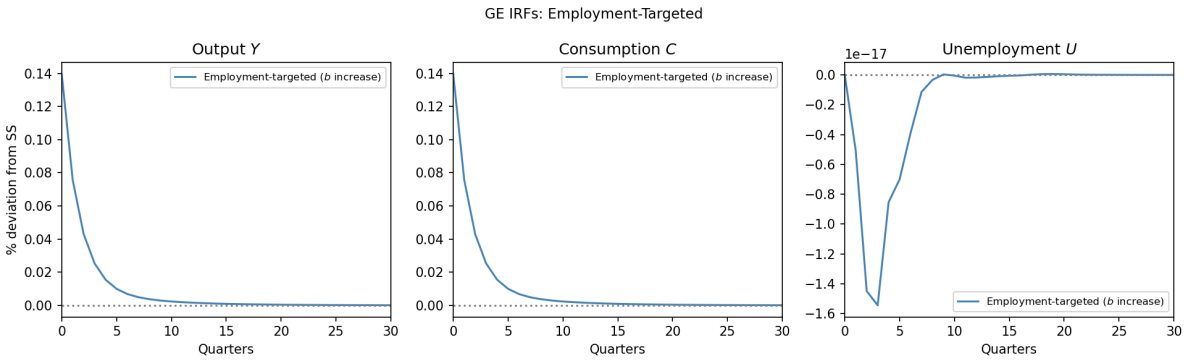

Figure 4: IRFs: Employment-Targeted (shock in b)

These results establish a baseline: an employment-targeted transfer achieves a well-identified direct income effect concentrated among high-MPC households, with limited GE amplification. The next step, comparing this to a cost-equivalent universal transfer, will reveal whether targeting dominates on aggregate output grounds, or whether the broader reach of universal transfers compensates for their lower average MPC.

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Appendix: Calibration Table

Table 1: Calibration

Parameter	Description	Value	Source / Target
Households			
σ	Inverse Risk Aversion	2.0	Standard in HANK literature
β	Discount factor	0.942	Estimated: asset market clearing
Labor market			
f	Job-finding probability (quarterly)	0.4	$U^* = s/(s+f) = 20\%$
s	Separation probability (quarterly)	0.1	$U^* = s/(s+f) = 20\%$
Idiosyncratic income process			
ρ_e	Persistence of log-productivity	0.966	Kaplan, Moll, and Violante (2018)
σ_e	Std. dev. of productivity innovation	0.5	Kaplan, Moll, and Violante (2018)
n_e	Number of Rouwenhorst nodes	7	Standard discretization
Asset grid			
$\underline{a}$	Borrowing constraint	0.0	No borrowing (McKay and Reis 2016)
$\bar{a}$	Maximum asset	200.0	Sufficient to cover top of dist.
n_a	Number of asset grid points	500	Accuracy near constraint
Prices and aggregate			
Z	Total factor productivity	1.0	Normalization
π	Steady-state inflation	0.0	Standard SS assumption
r	Real interest rate	0.02	2% annual average (US data)
w	Real wage	0.901	Estimated: NKPC ($w = Z/\mu$)
Y	Aggregate output	0.618	Estimated: goods market clearing
Government			
τ	Flat labor income tax rate	0.25	Kaplan, Moll, and Violante (2018)
b	Unemployment benefit (flat)	0.1	Shock variable of interest
B	Government bond supply	1.0	Normalization
Monetary policy			
r^*	Neutral real interest rate	0.02	Consistent with r at SS
ϕ	Taylor rule inflation coefficient	1.5	Standard (Taylor 1993)
Firms			
μ	Price markup	1.11	10% markup (Basu and Fernald 1997)
κ	NKPC slope (Rotemberg)	0.1	Hazell et al. (2022)

The estimated parameters (β, w, Y) are the solution to the system of assets and goods market and NKPC.

EIS $= 1/\sigma = 0.5$ is consistent with micro estimates surveyed by Valickova, Havranek, and Horvath (2015).